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Ekman Spirals with Variable Eddy Viscosity

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Contents

1 Background

1.1 Introduction

The Ekman spiral is a famous consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the atmosphere and ocean, the wind high up in the atmosphere, and current in the deep ocean, will not give rise to a flow in the same direction at the surface. The adjustment of the flow from the surface will thus spiral, and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. The surface flow is often at an angle from the flow aloft (or down under in the ocean) by an angle of more or less 45 ° [Ekman, 1905]. Due to this turning of the wind, the net flux of the wind will be perpendicular to the wind aloft (or current below). This is known as the Ekman transport and is crucial for studying both meteorology and oceanography.

However, this derivation with a surface angle of 45 ° assumes a constant viscosity in the medium. However, this is a simplification of reality, where a turbulent boundary layer is present. Thus, in reality the viscosity should vary with altitude. This would then vary the surface angle from 45 °, and thus also the Ekman transport. In this article, a varying viscosity with altitude will be studied under the Ekman equation. This will be done both theoretically and numerically and will also be compared to model data. This will give a deeper insight into the Ekman spiral and provide insight into when a surface angle different from 45 ° can be observed.

2 Theory

2.1 Problem formulation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming friction, incompressible flow, negligible vertical motion, gives the horizontal momentum equations as

$$\begin{cases} -fv = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial x} + \dfrac{\partial}{\partial z}\left(\nu \dfrac{\partial u}{\partial z}\right), & \text{(1a)} \\ fu = -\dfrac{1}{\rho}\dfrac{\partial p}{\partial y} + \dfrac{\partial}{\partial z}\left(\nu \dfrac{\partial v}{\partial z}\right), & \text{(1b)} \end{cases}$$

where u and v are the horizontal velocity components, f is the Coriolis parameter, ρ is the fluid density, and ν is the viscosity. Far away from the surface, shear flow and friction becomes negligible and the flow will thus approach geostrophic balance, $(u, v) \to (u_g, v_g)$, as were one can neglect the friction term in Equation 1b. Subtracting the geostrophic balance from Equation 1b thus gives

$$\begin{cases} -f(v - v_g) = \dfrac{\partial}{\partial z}\left(\nu \dfrac{\partial u}{\partial z}\right), & \text{(2a)} \\ f(u - u_g) = \dfrac{\partial}{\partial z}\left(\nu \dfrac{\partial v}{\partial z}\right). & \text{(2b)} \end{cases}$$

This problem can be formulated with one equation if one introduces a complex velocity defined as

$$U \equiv u + iv, \tag{3}$$

and a complex geostrophic velocity as $U_g = u_g + iv_g$. Using Equation 2b with the complex velocities U and U_g the coupled equations can be combined into the Ekman equation

$$\boxed{\frac{\partial}{\partial z}\left(\nu \frac{\partial U}{\partial z}\right) = if(U - U_g).} \tag{4}$$

With the boundary conditions

$$U(0) = 0, \tag{5a}$$
$$U(z \to \infty) = U_g. \tag{5b}$$

2.2 The classical solution

The classical solution to the Ekman equation is obtained when the viscosity is assumed to be constant with altitude, ν = constant, and can be found in Ekman [1905]. In this case he classical Ekman spiral solution from Equation 4 can obtained as

$$U(z) = U_g \left(1 - e^{-\frac{1+i}{\sqrt{2}}\frac{z}{h_{\mathrm{Ek}}}}\right), \tag{6}$$

where

$$h_{\mathrm{Ek}} \equiv \sqrt{\frac{\nu}{f}} \tag{7}$$

is the Ekman depth scale. One can note that by taking the argument of the change by altitude $\frac{\partial U}{\partial z}|_{z=0}$, one can measure the angle of Equation 6 as $45\,^\circ$.

2.3 The two layer model

The next natural step from a constant viscosity profile is to assume two layers of constant viscosity in each layer. One can label the lower level as having the viscosity ν_1 up until a height of H, where the next viscosity of ν_2 takes over. From this assumption one would obtain a superimposed solution similar to Equation 6 as

$$\begin{cases} U_1(z) = A_1 e^{\frac{1+i}{\sqrt{2}}\frac{z}{h_{\mathrm{Ek1}}}} + B_1 e^{-\frac{1+i}{\sqrt{2}}\frac{z}{h_{\mathrm{Ek1}}}} + U_g, & (8a) \\ U_2(z) = B_2 e^{-\frac{1+i}{\sqrt{2}}\frac{z}{h_{\mathrm{Ek2}}}} + U_g & (8b) \end{cases}$$

where the Ekman depth scales are defined for each layer like in Equation 7. To solve for the constants above one has to use the fact that the material flux has to be constant. This gives the jump condition as

$$U_1 = U_2 \quad \text{at } z = H, \tag{9a}$$
$$\nu_1 \frac{\partial U_1}{\partial z} = \nu_2 \frac{\partial U_2}{\partial z} \quad \text{at } z = H. \tag{9b}$$

Using this and the boundary conditions at Equation 5 one obtains a expression between for factors as

$$A_1 = -U_g \frac{1-\gamma}{1-\gamma+e^{\sqrt{2}(1+i)\varphi}(1+\gamma)}, \text{ where } \gamma = \sqrt{\frac{\nu_2}{\nu_1}} \tag{10a}$$
$$B_1 = -U_g \frac{(1+\gamma)\, e^{\sqrt{2}(1+i)\varphi}}{1-\gamma+e^{\sqrt{2}(1+i)\varphi}(1+\gamma)}, \tag{10b}$$
$$B_2 = -U_g \frac{2e^{\frac{1+i}{\sqrt{2}}\varphi\left(1+\sqrt{\frac{\nu_1}{\nu_2}}\right)}}{1-\gamma+e^{\sqrt{2}(1+i)\varphi}(1+\gamma)}, \tag{10c}$$

where φ is the non-dimensional height defined as

$$\varphi = \frac{H}{h_{\mathrm{EK1}}}. \tag{11}$$

The constant R is defined as

$$R = \frac{1-\gamma}{1+\gamma}. \tag{12}$$

Using this one can define the surface angle as

$$\theta = \arg \left.\frac{\partial U_1}{\partial z}\right|_{z=0} = 45^\circ + \arctan\left[\frac{2Re^{\sqrt{2}\,\varphi}\sin(\sqrt{2}\,\varphi)}{e^{2\sqrt{2}\,\varphi} - R^2}\right] \tag{13}$$

Observing Equation 13, and assuming that $\nu_1 > \nu_2$, one can see that the solution will always provide a result greater than $45\,^\circ$, except between the $\varphi \in \left(\frac{\pi}{\sqrt{2}}, \sqrt{2}\pi\right)$ were a angle below $45\,^\circ$ can be observed.

2.4 The exponential decaying viscosity

Another viscosity profile that can provide a fully analytical solution is an exponentially decaying viscosity with altitude. Here the viscosity is set to decay from an initial viscosity of ν_0, with a decay rate of k as $\nu(z) = \nu_0 \exp(-kz)$. To solve Equation 4 with this viscosity profile one can introduce two new parameters

$$t = 2\sqrt{i}\,\varphi\, e^{\frac{kz}{2}}, \quad U^* = \frac{U - U_g}{t}. \tag{14}$$

The nondimensional parameter φ was introduced above in this problem, just like in the 2-layer model, but here it is defined as

$$\varphi = \frac{1}{k\, h_{\mathrm{EK0}}} = \frac{1}{k}\sqrt{\frac{f}{\nu_0}}. \tag{15}$$

Using these newly defined variables Equation 4 transforms into

$$t^2\frac{\partial^2 U^*}{\partial t^2} - t\frac{\partial U^*}{\partial t} - t^2 U^* = 0, \tag{16}$$

which is a differential equation with a standard solution. The standard solution is a modified Bessel equation as

$$U^* = C_1 I_1(t) + C_2 K_1(t), \tag{17}$$

where C_1 and C_2 are constants, while $I_1(t)$ and $K_1(t)$ are well-defined modified Bessel functions of the first and second kind of order one, respectively. By applying the previously defined boundary conditions from Equation 5 one can find the two constants. Applying the geostrophic boundary condition from Equation 5b one can see that $C_1 = 0$ since $I_1(t)$ does not go to zero when $t \to \infty$ (which occurs when $z \to \infty$). Applying the surface boundary condition from Equation 5a one can find an expression for the second constant. Transforming the solution to the normal complex velocity as in Equation 3 one can obtain the solution

$$U(z) = U_g\left[1 - \frac{e^{\frac{kz}{2}} K_1\left(2\sqrt{i}\,\varphi\, e^{\frac{kz}{2}}\right)}{K_1\left(2\sqrt{i}\,\varphi\right)}\right]. \tag{18}$$

Using Equation 18 one can derive the surface angle of this model as

$$\theta = \arg \left.\frac{\partial U}{\partial z}\right|_{z=0} = \arg\left[U_g\sqrt{i}\,\frac{K_0\left(2\sqrt{i}\,\varphi\right)}{K_1\left(2\sqrt{i}\,\varphi\right)}\right], \tag{19}$$

where K_0 is the modified Bessel function of the second kind of order zero.

2.5 The 1.5 layer model

In the previous sections two completely analytical solutions for two different altitude-varying viscosity profiles were derived. The difficulty in solving Equation 4 comes from the altitude dependence of the viscosity profile $\nu(z)$. This introduces variable coefficients into the complex second-order ordinary differential equation, which in general prevents closed-form analytical solutions. Analytical solutions therefore exist only for specific viscosity profiles that allow the equation to be transformed into a solvable form.

Equation 4 is furthermore also difficult to solve in a numerical setting, as the upper boundary condition is a limit. This will put limits on the viscosity profiles one can study, as the viscosity cannot increase with altitude continuously. In the physical world, this makes sense, since we assume that high up in the atmosphere (or deep down in the ocean), friction and shear stress become negligible. This was what was labelled as the geostrophic limit. However, by limiting ourselves to only upper-level decreasing viscosity profiles, the differential equation becomes more difficult to solve. Furthermore, if one wants to solve for a dimensionless height parameter, too complex viscosity profiles become hard to deal with.

A solution to this problem is to introduce the 1.5 layer model, which is like the 2-layer model, except that we assume that the upper-level viscosity is negligible $\nu_2 \approx 0$. This also makes physical sense, since we know that both the atmosphere and the ocean consist of a boundary layer where turbulent mixing takes place. Reducing the problem to a 1.5 model will thus make it possible to test viscosity profiles that do not converge at $z \to \infty$, since the new domain is between $z \in [0, H]$. This modifies the boundary conditions to Equation 4 as

$$U(0) = 0, \tag{20a}$$

$$\left.\frac{\partial U}{\partial z}\right|_{z=H} = 0. \tag{20b}$$

Above, H becomes the height of the boundary layer, and the upper boundary condition was argued from the jump condition in the two-layer model, as seen in Equation 9b.

2.5.1 Simple viscosity profiles for the 1.5 layer model

Using the 1.5 layer model one can study some simple viscosity profiles to be evaluated numerically. The easiest, except for the constant viscosity, is a linearly varying viscosity with height as

$$\nu_\uparrow(z) = \frac{\nu_{\max}}{H} z, \tag{21a}$$

$$\nu_\downarrow(z) = \frac{\nu_{\max}}{H}\left(H - z\right), \tag{21b}$$

where $\nu_{\max}$ is the maximal viscosity. For both of these profiles one can define a dimensionless height coordinate as

$$\tilde{z} = \frac{z}{H} \implies \begin{cases} \nu_\uparrow(z) = \nu_{\max}\tilde{z} \\ \nu_\downarrow(z) = \nu_{\max}\left(1 - \tilde{z}\right). \end{cases} \tag{22}$$

One must note that this new coordinate changes the previous z-derivative, as a factor of $1/H$ appears when taking the derivative with respect to $\tilde{z}$. Using these linear viscosity profiles, together with the new dimensionless height coordinate, one can rewrite Equation 4 for the two profiles as

$$\frac{\partial}{\partial \tilde{z}}\left(\underbrace{\tilde{z}}_{\equiv\tilde{\nu}_\uparrow}\frac{\partial U}{\partial \tilde{z}}\right) = i\underbrace{\frac{Hf^2}{\nu_{\max}}}_{=\varphi^2}(U - U_g), \tag{23a}$$

$$\frac{\partial}{\partial \tilde{z}}\left(\underbrace{[1-\tilde{z}]}_{\equiv\tilde{\nu}_\downarrow}\frac{\partial U}{\partial \tilde{z}}\right) = i\underbrace{\frac{Hf^2}{\nu_{\max}}}_{=\varphi^2}(U - U_g). \tag{23b}$$

Above the dimensionless height φ was again introduced following the logic from before.

Another simple profile to evaluate is a parabolic increasing viscosity as

$$\nu_p = \frac{4\nu_{\max}}{H^2}\left(H - z\right)z. \tag{24}$$

Again, using the non-dimensional height coordinate $\tilde{z}$ and the dimensionless height φ, one can rewrite Equation 4 as

$$\frac{\partial}{\partial \tilde{z}}\left(\underbrace{4\tilde{z}\,[1-\tilde{z}]}_{\equiv\tilde{\nu}_p}\frac{\partial U}{\partial \tilde{z}}\right) = i\underbrace{\frac{Hf^2}{\nu_{\max}}}_{=\varphi^2}(U - U_g). \tag{25}$$

2.6 The Ekman transport

For a theoretically or numerically derived solution, one can derive the Ekman transport, which may be a more useful quantity. The Ekman transport is defined as the mass flux per meter for the entire Ekman spiral. In the classical solution the Ekman transport is $90\,^\circ$ to the right of the surface flow. However one must note that this angle is obtained by subtracting the Ekman transport relative to the geostrophic flow o the surface flow, i.e $135\,^\circ{-}45\,^\circ{=}90\,^\circ$ in classical theory. However, for an obtained solution $U(\tilde{z},\varphi)$ as a function of the dimensionless height and layer thickness, one can obtain the mass flux as

$$\Phi_T(\varphi) = \int_0^1 \left[U(\tilde{z},\varphi) - U_g\right] d\tilde{z}. \tag{26}$$

To obtain the transport angle one can simply compute the argument as

$$\theta_T(\varphi) = \arg\left[\Phi_T(\varphi)\right]. \tag{27}$$

Thus the relative angle between the surface transport and the upper geostrophic flow can be calculated as

$$\theta_R = \theta_T - \theta. \tag{28}$$

3 Methodology

3.1 Numerical method

For arbitrary viscosity profiles, Equation 4 generally does not admit a closed-form analytical solution. The equation was therefore solved numerically as a boundary value problem using the `solve_bvp` routine available in `SciPy` [Virtanen et al., 2020]. However, it was previously shown that the equation could be rewritten for a dimensionless layer thickness φ. Thus, using an arbitrary height-normalized height-dependent eddy viscosity profile, $\tilde{\nu}(z)$, one can write a simpler differential equation. Separating the equation into the zonal and meridional velocity components, one obtains

$$\frac{\partial}{\partial z}\left(\tilde{\nu}(z)\frac{\partial u}{\partial z}\right) = -\varphi^2 v, \tag{29a}$$

$$\frac{\partial}{\partial z}\left(\tilde{\nu}(z)\frac{\partial v}{\partial z}\right) = \varphi^2 (u - u_g), \tag{29b}$$

where the wind aloft was set as a purely zonal wind u_g. To apply the numerical boundary value solver, the equations were rewritten as a first-order system by introducing the new variables

$$u' = \frac{\partial u}{\partial z}, \qquad v' = \frac{\partial v}{\partial z}. \tag{30}$$

Thus one can expand Equation 29b into four coupled first-order equations

$$\begin{cases} \frac{\partial u}{\partial z} &= u', \\ \frac{\partial u'}{\partial z} &= \frac{-\varphi^2 v - \tilde{\nu}' u'}{\tilde{\nu}}, \\ \frac{\partial v}{\partial z} &= v', \\ \frac{\partial v'}{\partial z} &= \frac{\varphi^2 (u-u_g) - \tilde{\nu}' v'}{\tilde{\nu}}, \end{cases} \tag{31}$$

with the boundary conditions

$$u(\varphi = 0) = 0, \qquad u'(\varphi = 1) = 0, \tag{32a}$$

$$v(\varphi = 0) = 0, \qquad v'(\varphi = 1) = 0. \tag{32b}$$

The equations were solved iteratively on a vertical grid. The solver required an initial guess, which was chosen based on the initial setup of the system. The initial guess was therefore set as the classical solution for a constant viscosity. The final solution was found when the iterative solution reached a steady state for the differential equations with the imposed boundary conditions.

4 Results

4.1 The theoretical models

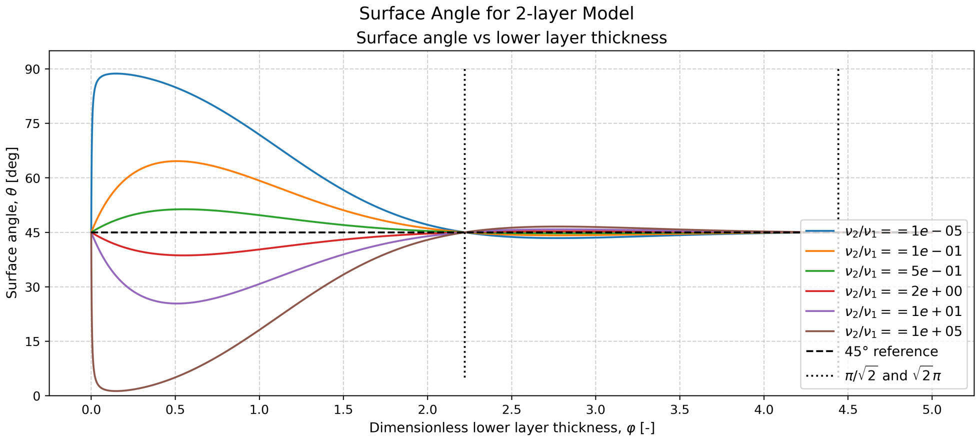

Figure 1: The surface angle as a function of the dimensionless layer thickness $\varphi = H/h_{\mathrm{EK1}}$ for the two-layer viscosity model. Different coloured curves correspond to different layer viscosity ratios. The classical Ekman angle of 45 $^\circ$ is shown as a reference.

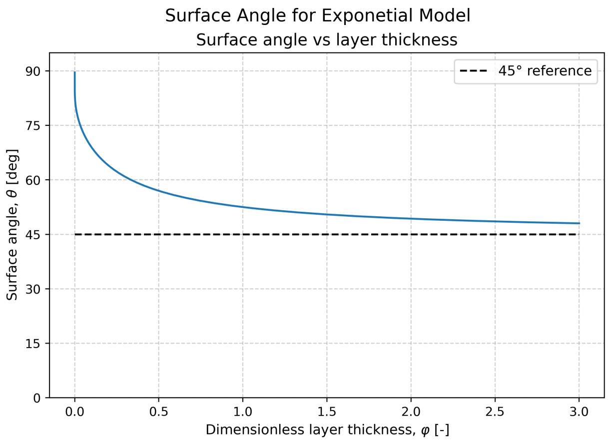

Figure 2: Surface angle for the exponentially decaying viscosity model as a function of the dimensionless parameter $\varphi = H/h_{\mathrm{EK}}$. The solution assumed a purely zonal geostrophic flow.

4.2 The numerical 1.5 model

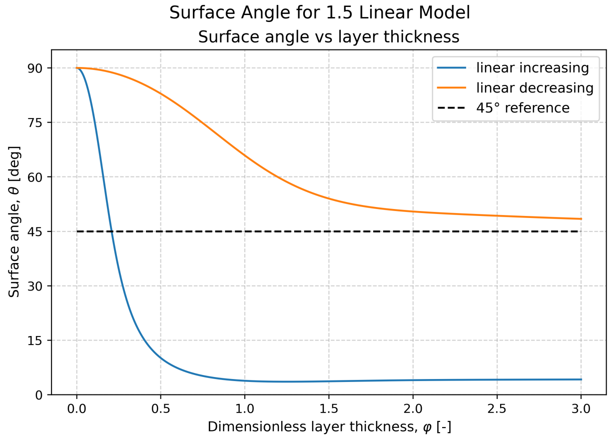

Figure 3: Numerically computed surface angles for linearly increasing and decreasing viscosity profiles in the 1.5-layer model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{EK}}$. The solution assumed a purely zonal geostrophic flow.

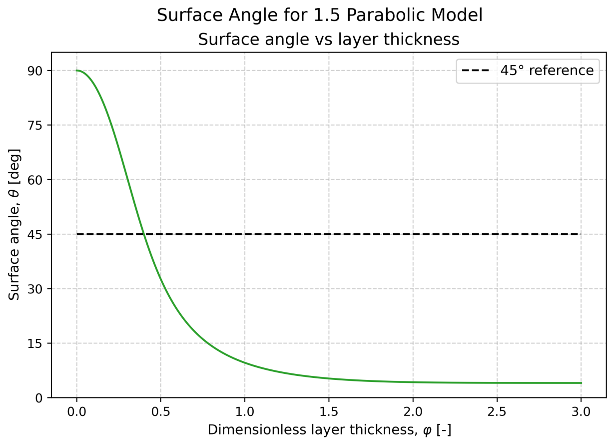

Figure 4: Numerically computed surface angles for a parabolic viscosity profile in the 1.5-layer model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{EK}}$. The solution assumed a purely zonal geostrophic flow.

4.2.1 The Ekman transport

One interesting observation in Figure 5–8 is that if one computes the relative transport angle, i.e. the angle between the surface angle and the transport angle, one obtains a relative value of $90\,^{\circ}$. Thus, these models give the same Ekman transport angle as the classical Ekman theory.

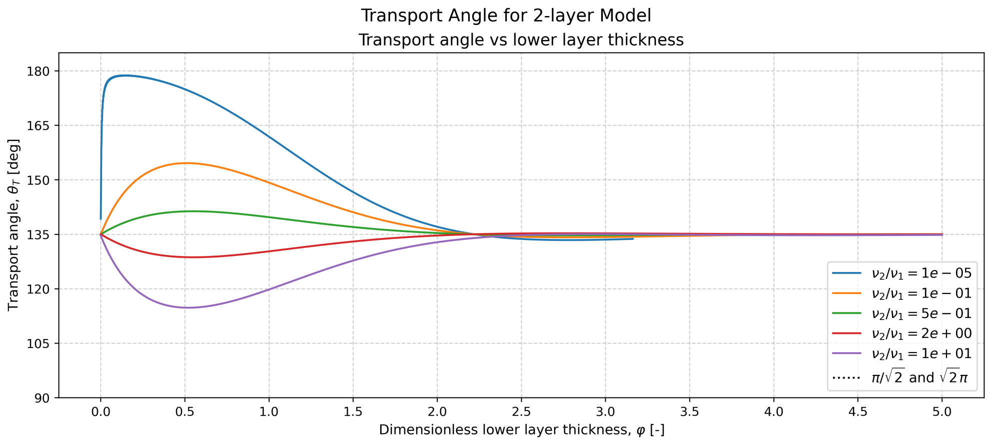

Figure 5: Numerically computed transport angles for the two layer viscosity profile. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{EK}}$. Different coloured curves correspond to different layer viscosity ratios. The solution assumed a purely zonal geostrophic flow.

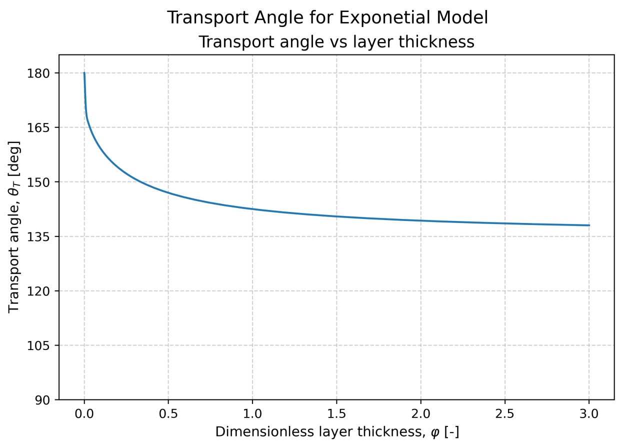

Figure 6: Numerically computed transport angles for the exponential viscosity profile. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{EK}}$. The solution assumed a purely zonal geostrophic flow.

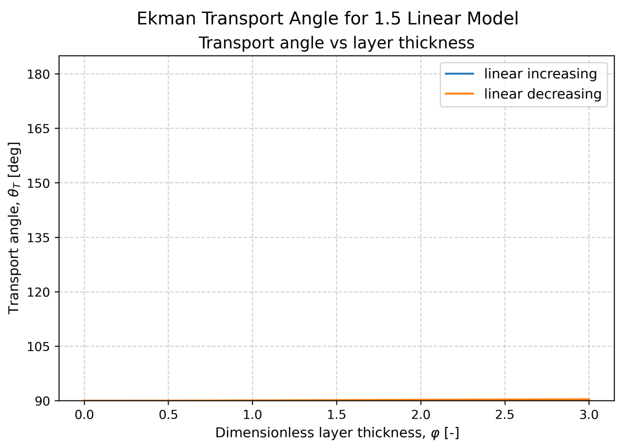

Figure 7: Numerically computed transport angles for linearly increasing and decreasing viscosity profiles in the 1.5-layer model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{EK}}$. The solution assumed a purely zonal geostrophic flow.

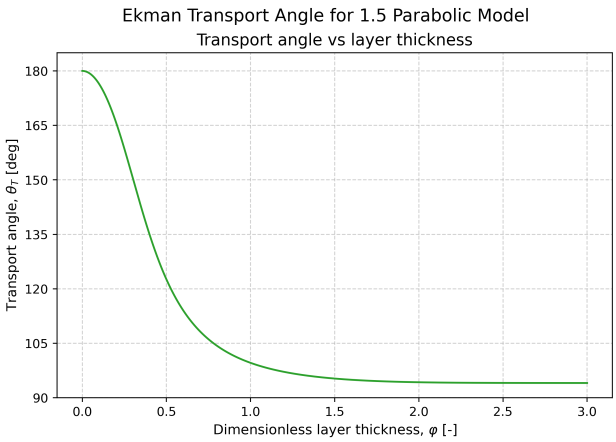

Figure 8: Numerically computed transport angles for a parabolic viscosity profile in the 1.5-layer model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{EK}}$. The solution assumed a purely zonal geostrophic flow.

5 Discussion

6 Conclusion

A The Source Code

The source code for this article can be found on GitHub under:

`https://github.com/FredrikBergelv/Ekman-Spiral-Project`

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